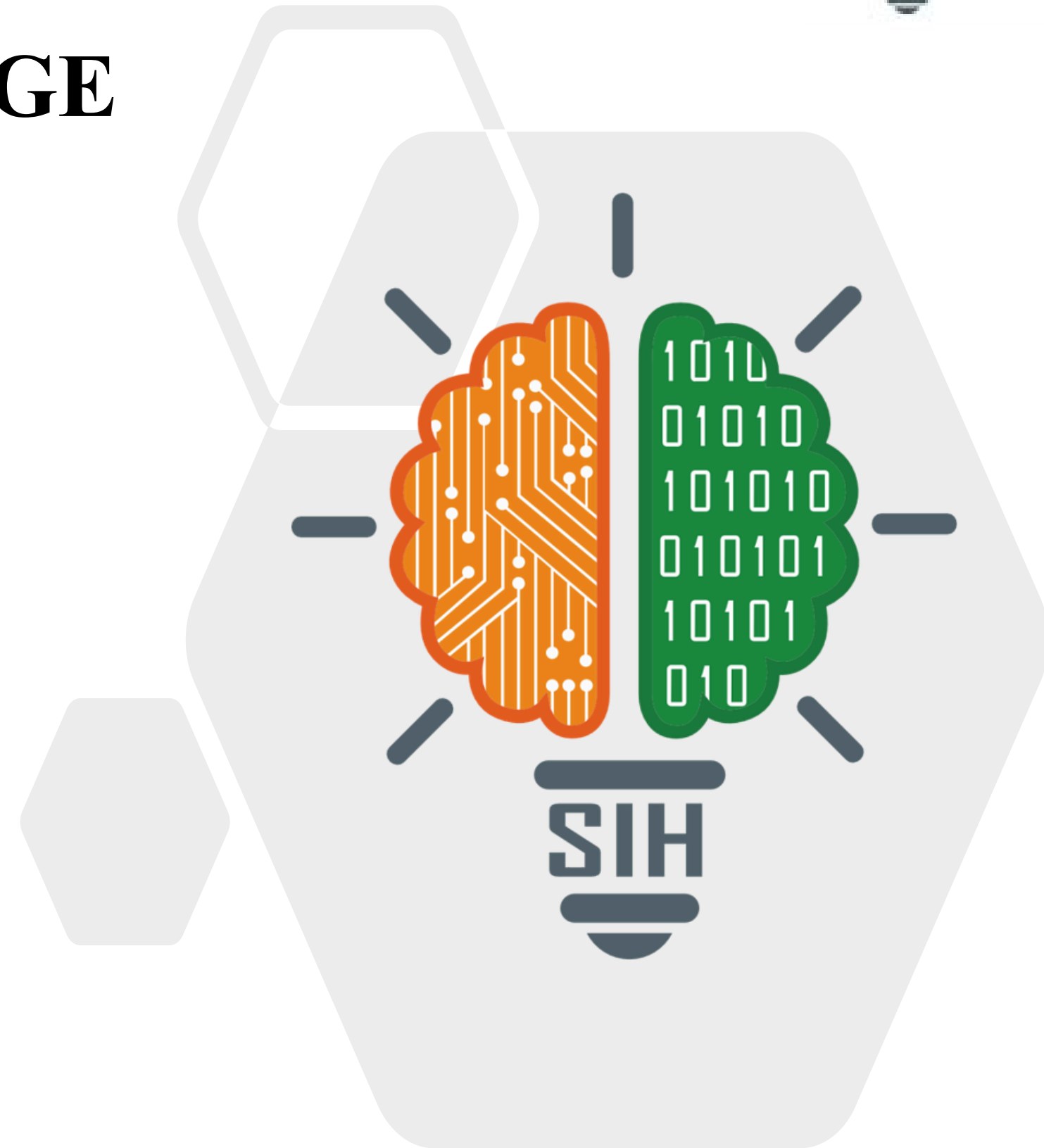


TITLE PAGE

- **Problem Statement ID** : SIH26104
- **Problem Statement Title** : AI-Powered
Real-Time Detection and Prevention of
Voice Cloning Impersonation Attacks
- **Theme** : Blockchain & Cybersecurity
- **PS Category** : Software
- **Team Name** : First and Final Attempt



PROBLEM ADDRESSED

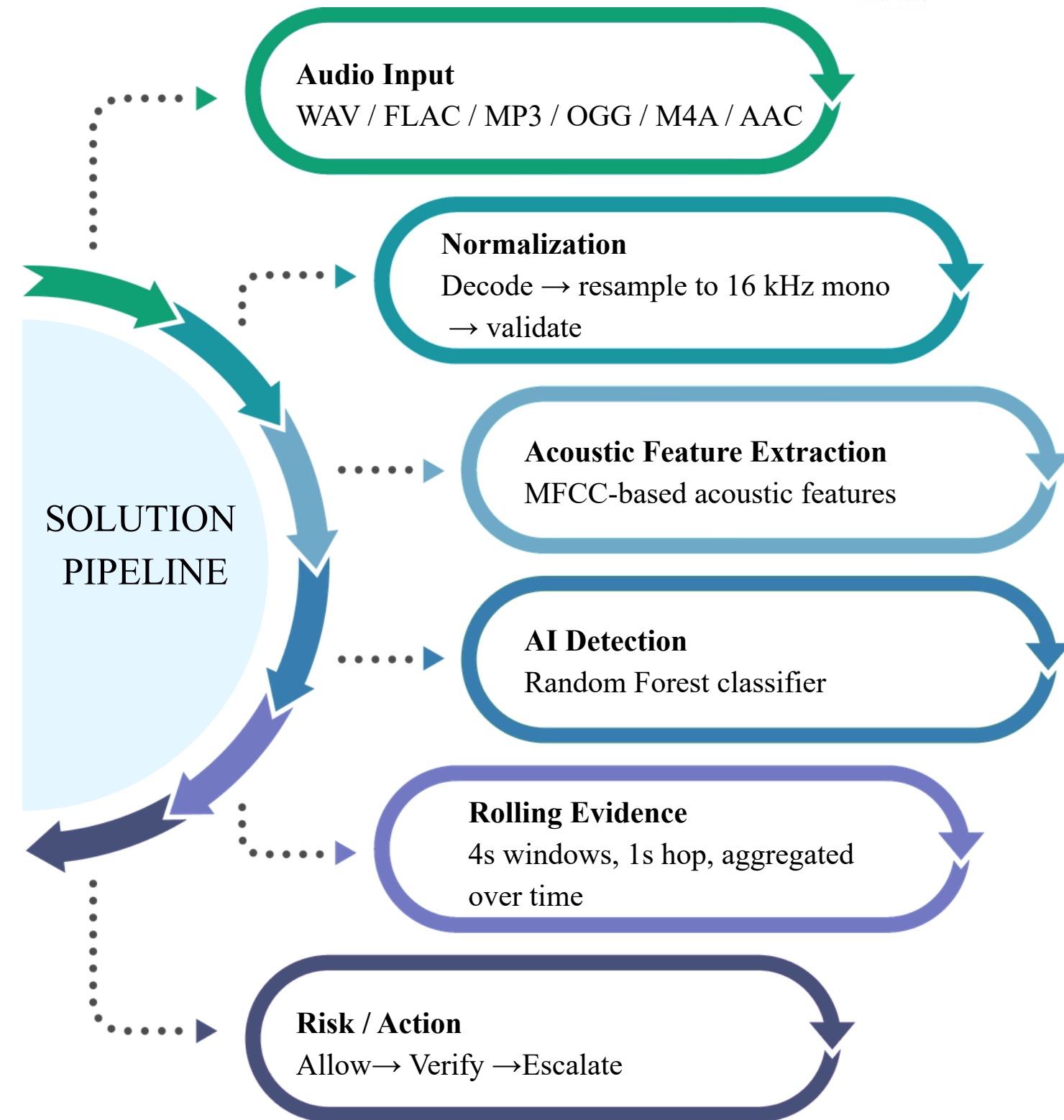
- Convincing synthetic voices enable impersonation
- Real-world audio introduces uncertainty
- Voice-based security needs more than a single binary decision

PROPOSED SOLUTION

- Adds a security-oriented layer around a voice-cloning classifier
- Analyzes rolling, overlapping 4-second audio windows
- Fuses recent evidence into a proportionate Allow / Verify / Escalate action

INNOVATION

- Adds a security-oriented layer around a voice-cloning classifier
- Analyzes rolling, overlapping 4-second audio windows
- Fuses recent evidence into a proportionate Allow / Verify / Escalate action



First and Final Attempt

TECHNOLOGY STACK

Python

Streamlit

FastAPI

scikit-learn

WebSocket

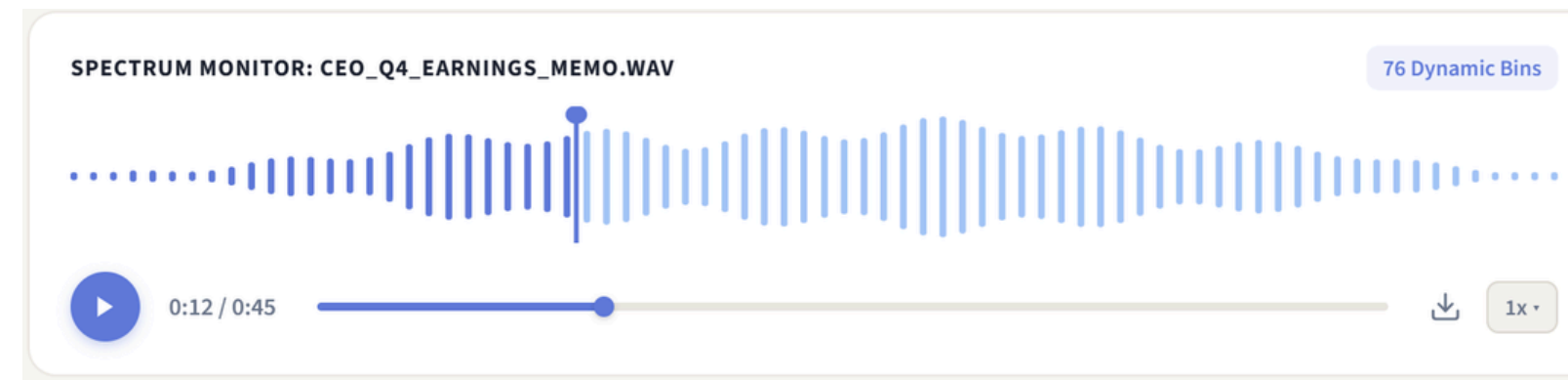
Librosa

NumPy

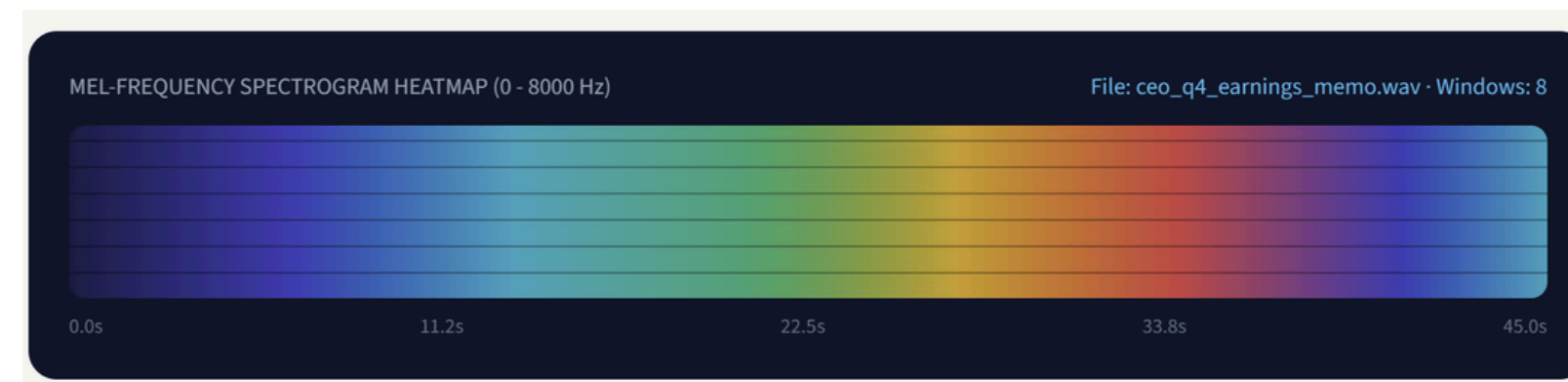
TECHNICAL APPROACH

DETECTION EVIDENCE

Waveform



Spectrogram



Acoustic Features (MFCC-based)

- Original detector: 13 mean MFCC features
- Improved prototype adds: MFCC mean/std, delta & delta-delta, spectral centroid, bandwidth, rolloff, zero-crossing rate, RMS energy, spectral contrast
- Prediction is treated as evidence, aggregated across rolling windows — not a one-shot verdict

FEASIBILITY AND VIABILITY



TECHNICAL

- Working web prototype end to end
- Python + Streamlit dashboard + FastAPI backend
- Random Forest classifier over acoustic features
- Modular audio processing pipeline

OPERATIONAL

- Supports common audio upload formats
- Prediction with waveform / spectrogram visualization
- Rolling-window analysis over time

PRIVACY

- Temporary audio processing only
- Metadata-only evidence returned
- No raw audio written to application event logs

CHALLENGES → MITIGATION

Noise / audio compression

Rolling windows smooth transient noise

Unseen synthesis techniques

Temporal evidence aggregation across windows

False positives

Independent verification for high-risk cases

Streaming latency

Future streaming optimization

Dataset / evaluation limitations

Robust, documented evaluation protocol

First and Final Attempt

IMPACT AND BENEFITS



IMPACT AREAS

SOCIAL

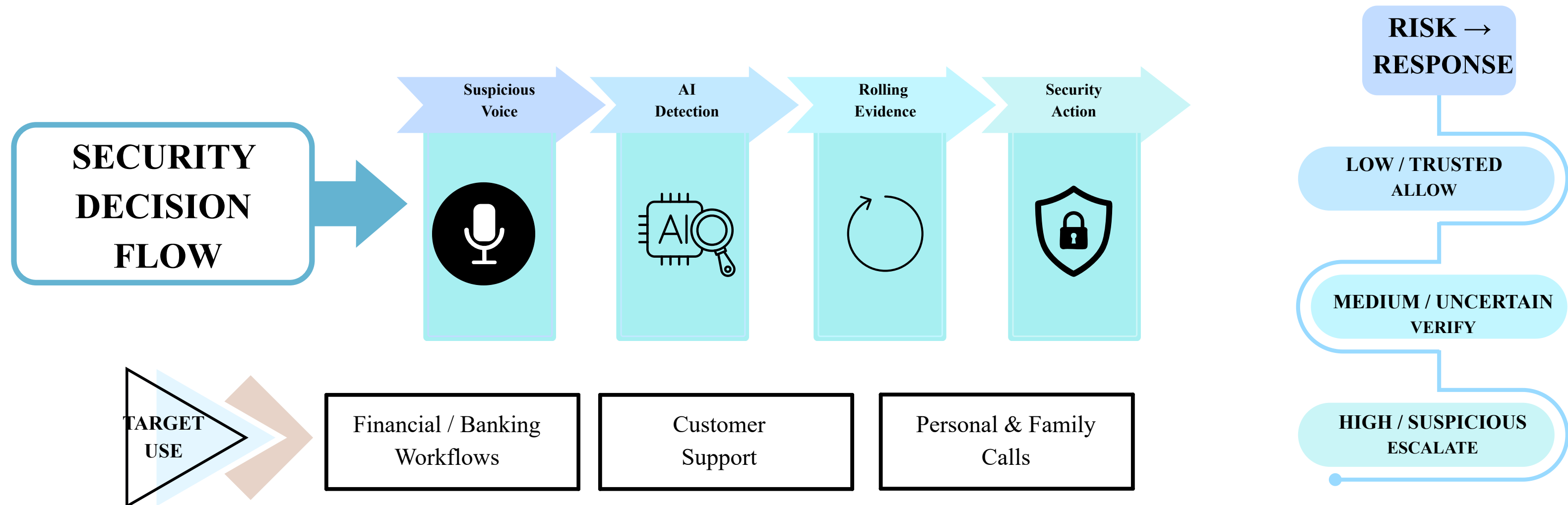
Protect people from voice scams and impersonation

ECONOMIC

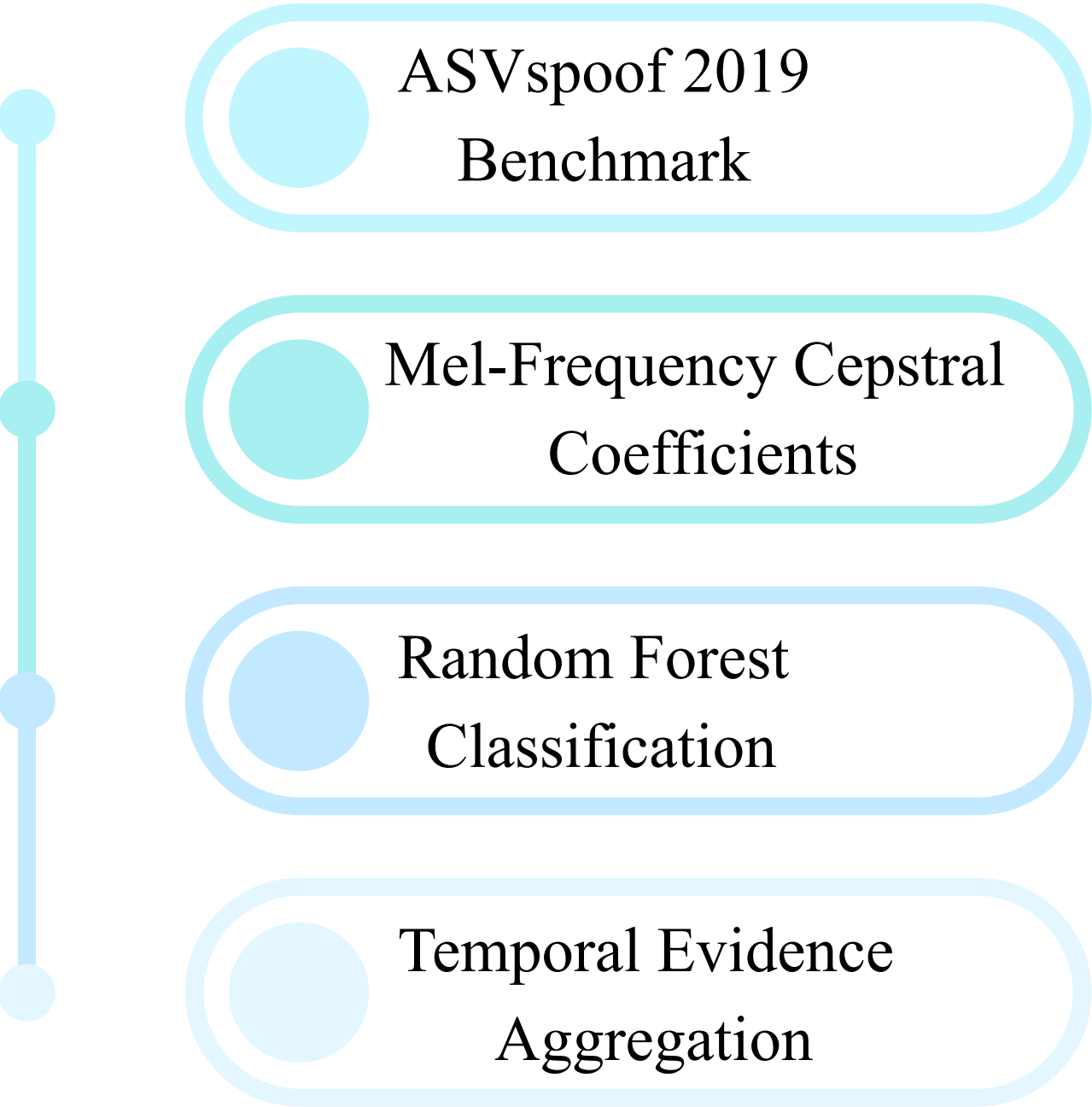
Reduce exposure to fraudulent approvals & risky interactions

SECURITY

Add AI-assisted evidence to voice security workflows



RESEARCH



FUTURE ROADMAP

CURRENT PROTOTYPE



FUTURE	Larger, more diverse anti-spoofing datasets
FUTURE	Pretrained speech embeddings
FUTURE	Dedicated OOD (out-of-distribution) detector
FUTURE	Audit-ready monitoring & robust production deployment